

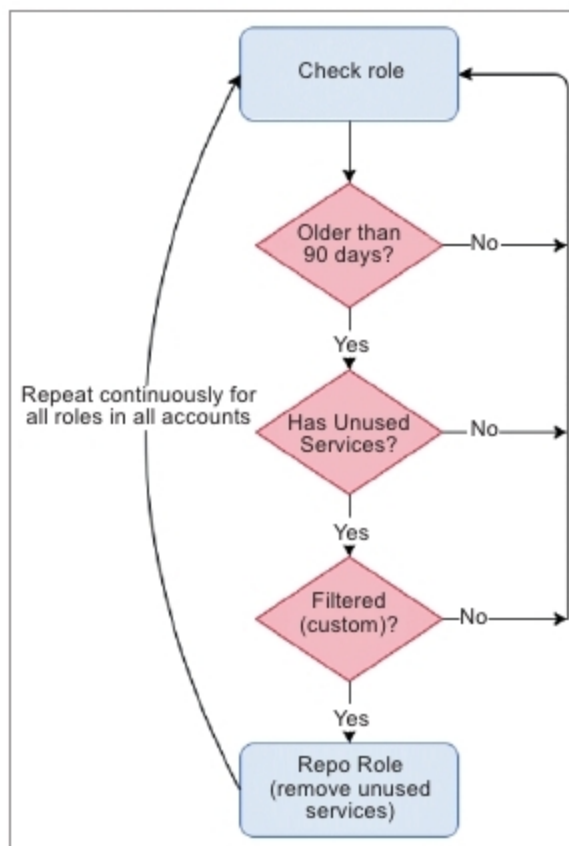


Figure 2: The Aardvark learning process

Terraform is platform-agnostic and helps you to deploy coding principles like automated tests, source code control, etc. It has a massive community backing, and many organisations have started using it to manage their infrastructure.

You can do a dry run of Terraform, which simulates actual changes without committing, a feature that is very useful for developers. Terraform Resource Graph creates a graphical view of all the resources, including the dependency between each resource. Terraform leverages the respective cloud provider's API to manage the infrastructure, which reduces the requirement for additional resources.

Terraform is a declarative language that describes the target level of the resource. Plugins are available for popular cloud vendors to manage the respective cloud resources. In addition, you can create your own plugins in many languages. Terraform mainly supports the Go language.

```

resource "aws_s3_bucket" "b" {
  bucket = "s3-mywebsite.com"
  acl    = "public-read"
  policy = "${file("mypolicy.json")}"
}
  
```

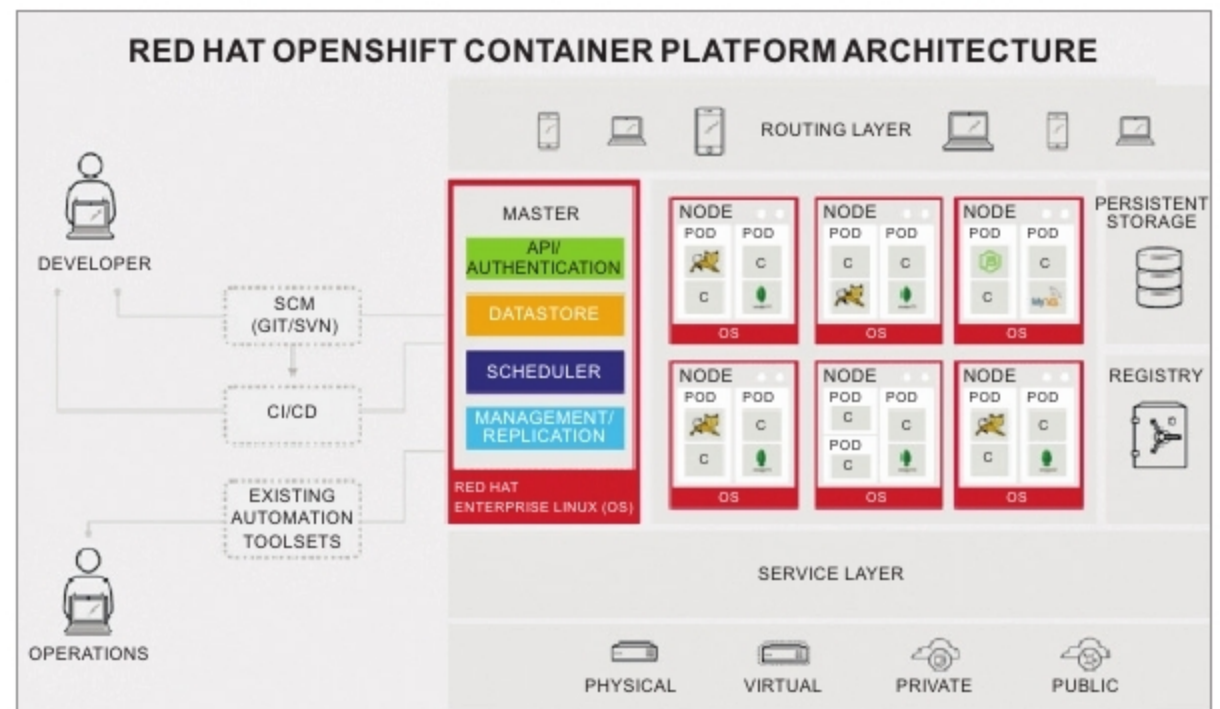


Figure 3: OpenShift container platform architecture

```

website {
  index_document = "index.html"
  error_document = "error.html"

  routing_rules = <<EOF
  [{
    "Condition": {
      "KeyPrefixEquals": "doc/"
    },
    "Redirect": {
      "ReplaceKeyPrefixWith":
        "document/"
    }
  }]
  EOF
}
  
```

git-secret

A usual developer practice is to encrypt the passwords/credentials used in the code and then save them in some secure place. We have multiple tools for this purpose, Vault being one of them. git-secret is a simple tool to store the secrets in the Git repo. git-secret uses *gpg* for encryption and decryption of secrets.

Use cases for keeping secrets are:

- To manage keys at a centralised location for encryption and decrypting
- Version control the secrets for better management
- Access control and leverage who can

encrypt and decrypt the secret

- No secrets to be stored in plain text
You can use git-secrets to store Oauth keys, DB passwords, application keys and other secret keys which are stored in plain text.

git-secret is a tool to store private data inside a Git repository. It encrypts the password using *gpg* and tracks files with the public keys of all the users who are already trusted. So users can decrypt the files using their respective personal secret keys only, and there will be no passwords that change here. When people leave the organisation or are replaced, just delete the public key, re-encrypt the files and they won't be able to decrypt secrets anymore.

Installing git-secret is very easy; just get the *gpg* key ready for the repo and install the secret with that key. The key config variables are:

- `$SECRETS_VERBOSE`: Sets verbose flag for all git-secret commands
- `$SECRETS_GPG_COMMAND`: Sets *gpg* alternatives (defaults value is *gpg*)
- `$SECRETS_EXTENSION`: Sets the secret file extension (default extn is *.secret*)
- `$SECRETS_DIR`: Sets the location where the files are to be stored (default folder is *.gitsecret*)